

Material Science

Composite Materials: Bridging Theory, Experiment and Innovation

Extra Studies (Basic FEA Study)

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Contents

List of Figures	2
List of Tables	2
FEA Simulation	2
1. Introduction	2
2. Missing Property Data	3
3. Simulation Setup	5
4. Simulation Results	8
5. Conclusion	8
6. References	9

List of Figures

Figure 1 - Graph showing a stress-strain graph of various materials	3
Figure 2 - FIA front wing testing setup [4]	5
Figure 3 - Material properties used in simulation.....	6
Figure 4 - 2026 Front Wing Simulation Setup	6
Figure 5 - Mesh details.....	7
Figure 6 - Mesh applied to part.....	7
Figure 7 - Visualised Results	8

List of Tables

Table 1 - Summary of estimated properties	4
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FEA Simulation

1. Introduction

This report follows on from the main body of the report which comes along side my submission of this report.

Formula One (F1) front wings must balance structural integrity whilst maintaining minimal weigh and optimal flexibility to comply with Fédération Internationale de l'Automobile (FIA) regulations. This small study investigates the mechanical performance of a 2026 F1 front wing under the FIA front wing deflection test setup. Using Finite Element Analysis (FEA) package on SolidWorks 2020, it will replicated similar environments that the front wing will be exposed to. The main objective of this report is to ensure the selected composite (M40J/Epoxy) will meet the deflection limit set by the FIA.

Finite Element Analysis (FEA) is useful for this study as is gives a simulation of complex geometries and conditions. It allows for quite accurate predictions of stress, strains, and deflections. This is very useful for models that will cost a lot to manufacture as it eliminates the risk of prototypes not working and spending lots of money on them.

The hypothesis is that the material will be able to pass the deflection test.

2. Missing Property Data

The material data for the M40J/Epoxy composite includes tensile strength and tensile modulus but doesn't specify Yield strength or Poisson's ratio of the material. Using standard practise both Yield strength and Poisson's ratio can be estimated. In this case tensile strength will be used as the ultimate tensile strength (UTS)

Yield Strength Estimation

Carbon fibre/epoxy composites undergo brittle failure with little to no plastic deformation, meaning a true yield point is difficult to find, this can be seen in figure 1 as the red line. [1] For simulation purposes, a value based on a fraction of the UTS will be used. A commonly used factor in materials is:

$$\sigma_y \approx 0.6 \times UTS$$

Given: Ultimate Tensile Strength (UTS) = 2500MPa

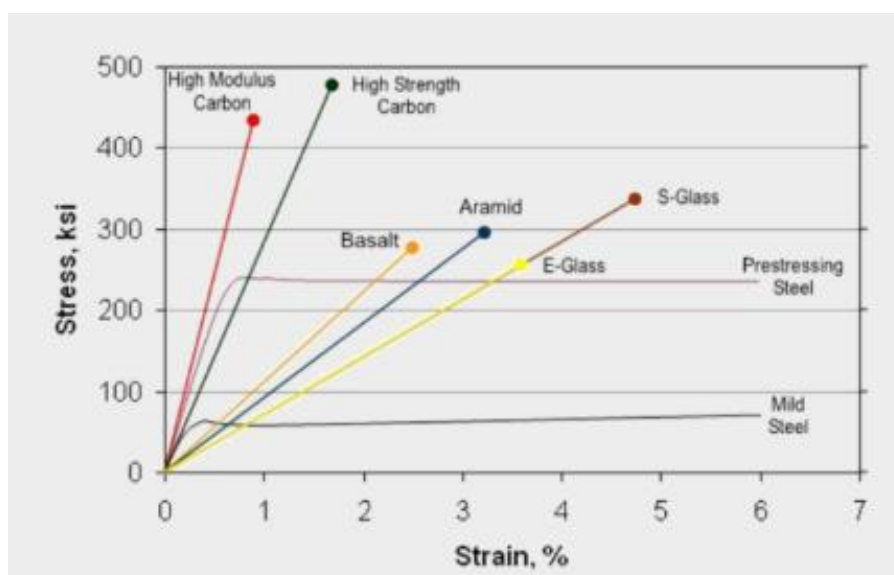
Calculation: Estimated Yield Strength = $0.6 \times 2500 = 1500\text{MPa}$

This can be seen in composite material handbooks like MIL-HDBK-17[2], this application is suitable for materials that fail with little to no plastic deformation.

Assumptions:

- Composite doesn't go into plastic deformation.

Figure 1 - Graph showing a stress-strain graph of various materials



Poisson's Ratio Estimation

Poisson's ratio (ν) is defined as:

$$\nu = \frac{E}{2G} - 1$$

Where:

- E = Longitudinal modulus
- G = Shear modulus

Since (G) was not in the table, the calculation cannot be done. Instead, using M40J/Epoxy material data can be compared to other carbon fibres and their Poisson's Ratio can be used. [3] Values for similar high modulus carbon/epoxy composites ~0.30. [4]

Estimated Poisson's Ratio ≈ 0.30

Assumptions:

- Based on published composite property databases for high modulus carbon fibres

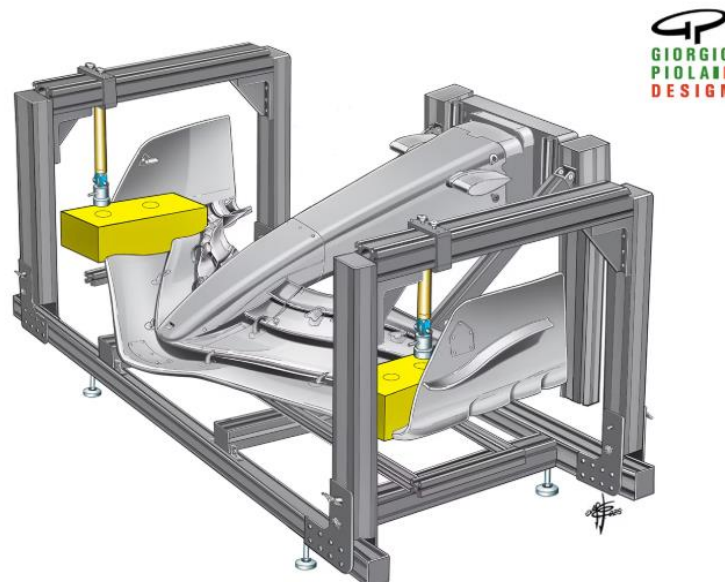
Table 1 - Summary of estimated properties

Property	Estimated Value
Yield Strength	1500 MPa
Poisson's Ratio	0.30

3. Simulation Setup

To complete the FEA simulation, SolidWorks 2020 was used. The 3D CAD model was modelled prior to use and follows the new 2026 FIA Formula One regulations. The simulation was based on the FIA testing process seen in figure 2. This is an illustration of how the FIA will test the Formula One front wings from 2025 onwards. As stated in Article 3.15.4 100kg will place on either side of the front wing and a maximum deflection of 10mm cannot be observed as it will fail the test. [5]

Figure 2 - FIA front wing testing setup [4]



A look at how front wing load tests are carried out by the FIA

To set the simulation up firstly the material must be fully defined. This simulation will use M40J/Epoxy composite. Using the technical data sheet from Toray [3], most properties can be entered, the ones that aren't present have been estimated using the processes explained in section 1. This data is entered under the property data, as seen in figure 3.

To simple the study I am using Linear Elastic Isotropic model type. Linear elastic means that the stress is proportional to the strain, and the model will return to its original shape once the load is removed. Isotropic means that all materials properties are equal in all directions. This simplifies the analysis as stiffness, and the deformation do not depend on orientation. [6] The study will also use maximum Von Mises stress as the failure criterion, again to simplify

the study. Von Mises criterion is used to predict yielding of materials. In this study not much yielding will occur. [7]

Figure 3 - Material properties used in simulation

Model Type:

Linear Elastic Isotropic

Save model type in library

Units:

SI - N/m^2 (Pa)

Category:

2026 Front Wing_Optimising_FE

Name:

M40J CF custom

Default failure criterion:

Max von Mises Stress

Description:

M40J CF custom

Source:

Sustainability:

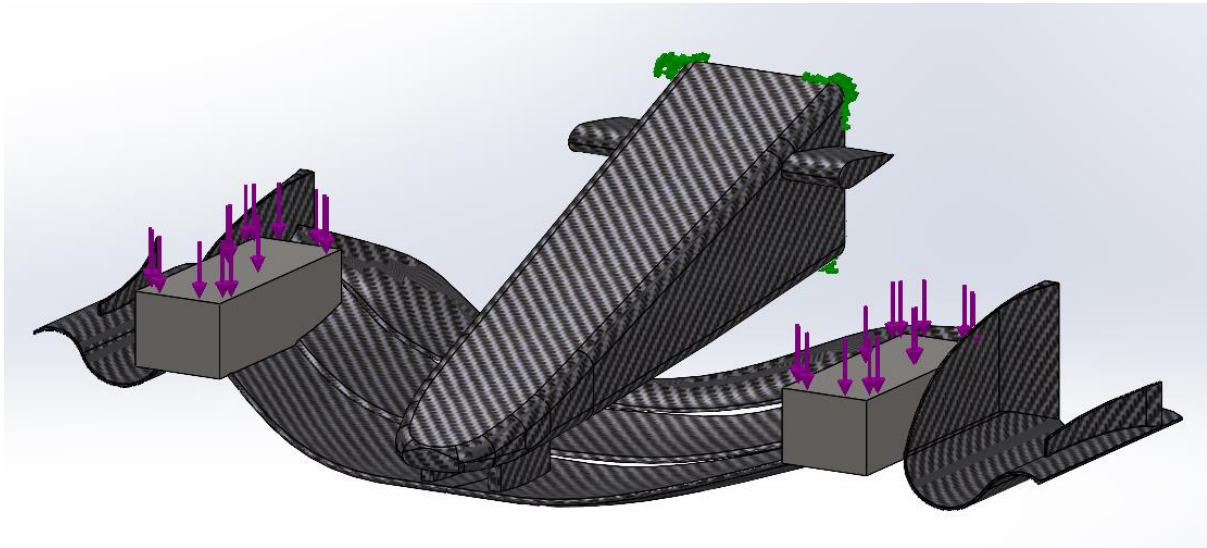
Undefined

Select...

Property	Value	Units
Elastic Modulus	2.26e+11	N/m^2
Poisson's Ratio	0.3	N/A
Shear Modulus		N/m^2
Mass Density	1750	kg/m^3
Tensile Strength	2500000000	N/m^2
Compressive Strength	1270000000	N/m^2
Yield Strength	1500000000	N/m^2

Once all the properties of the material are defined, the environmental setup can occur. The face that attaches the front wing to the chassis has been defined as a fixture and the top of the blocks have had 1000N applied to them (100kg $\approx$ 1000N), as seen in figure 4. I have tried to follow the illustration seen in figure 2 as closely as possible to ensure that the results are as similar to real testing as possible.

Figure 4 - 2026 Front Wing Simulation Setup

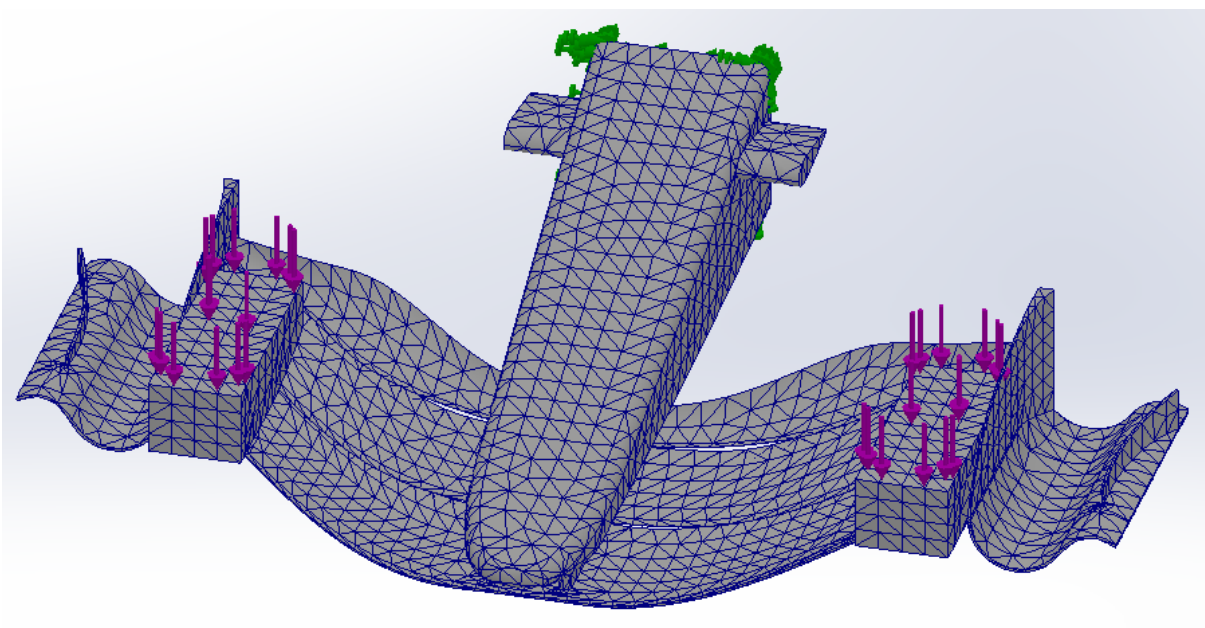


Once all the setup is completed a mesh can be created. Using a standard mesh in SolidWorks. The details of the mesh can be seen in figure 5, and when it is applied to the part in figure 6.

Figure 5 - Mesh details

Mesh Details	
Study name	Deflection (-Design_2-)
Mesh type	Solid Mesh
Mesher Used	Standard mesh
Automatic Transition	Off
Include Mesh Auto Loops	Off
Jacobian points for High quality mesh	16 points
Element size	39.0422 mm
Tolerance	1.95211 mm
Mesh quality	High
Total nodes	25078
Total elements	13858
Maximum Aspect Ratio	323.58
Percentage of elements with Aspect Ratio < 3	63.2
Percentage of elements with Aspect Ratio > 10	5.3
Percentage of distorted elements	0
Number of distorted elements	0
Time to complete mesh(hh:mm:ss)	00:00:08
Computer name	

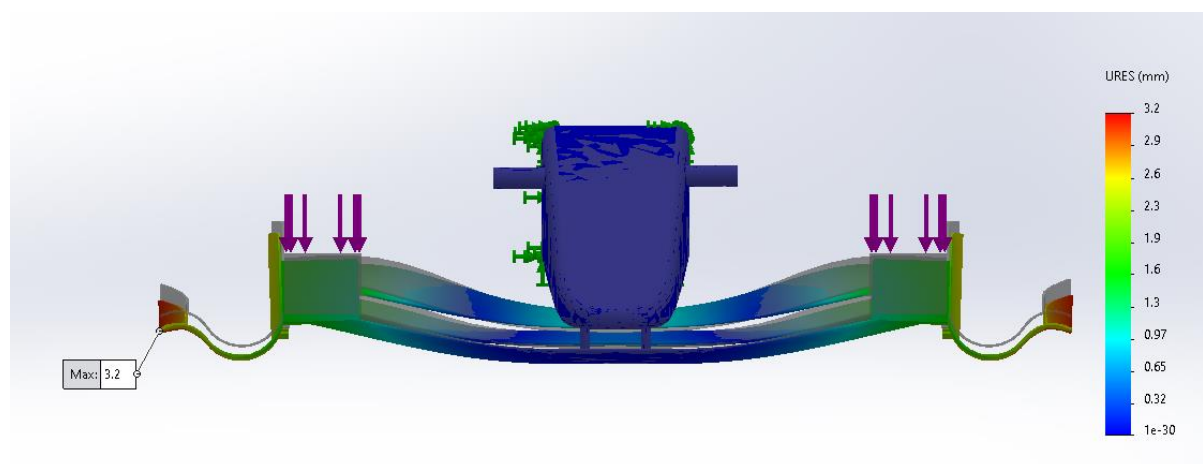
Figure 6 - Mesh applied to part



4. Simulation Results

Figure 7 illustrates the displacement distribution across the whole of the front under the specified conditions mentioned in section 2. The results show the maximum deflection occurs on the end plate and has a value of **3.2mm**. This can be seen as the red section on the contour in figure 7. The nose cone shows minimal deflection ($\sim 0\text{mm}$), as shown by the dark blue sections. It is important to examine the 3.2mm deflection still even though it is a lot under the max deflection, delamination or failure could occur here. The deflection results suggest that the front wing is strong enough to easily pass the FIA front wing deflection test.

Figure 7 - Visualised Results



5. Conclusion

The FEA simulation of the M40J/Epoxy composite 2026 F1 front wing shows a maximum deflection of 3.2mm under the loading conditions in the FIA testing conditions. This is much less the maximum deflection of 10mm. These results confirm that the chosen composite material can provide more than enough strength and stiffness to pass the deflection test.

This study has some limitation as it is only a basic study. The use of an isotropic material could show different results to what would be observed when using a real composite. Also, the mesh is not refined as the study is only being used to show rough results and required for compliance.

This simulation process has further developed my simulation skills and data analysis. Future work would include more complex FEA studies e.g. with a more refined mesh. Also real world conditions could be analysed using anisotropic material properties under real loading conditions.

6. References

- [1] 4.1.3. Fundamental Behavior of Carbon Fiber Epoxy Resin Composite Laminates [Internet]. Abbott Aerospace UK Ltd. 2019 [cited 2025 Nov 26]. Available from: <https://www.abbottaerospace.com/aa-sb-001/4-materials/4-1-composite-materials/4-1-3-fundamental-behavior-of-carbon-fiber-epoxy-resin-composite-laminates/>
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